

AI Website Chatbot Platform - Detailed Development Guide

Project Overview

The goal is to create a reusable AI chatbot platform that can be integrated into any website using a simple JavaScript widget.

The chatbot should act like a virtual employee that can:

1. Answer questions from website content.
 2. Search products and information through APIs.
 3. Read uploaded documents.
 4. Perform actions through APIs.
 5. Understand natural language instead of relying on predefined keywords.
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Core Concept

The chatbot should not simply search text.

Instead, it should:

- Understand the user's intent.
- Decide where information should come from.
- Fetch relevant data.
- Generate a natural response.

Example:

User: "Show me red shirts below ₹1000"

The chatbot should understand:

Intent:

- Product Search

Filters:

- Category = Shirt
- Color = Red
- Max Price = 1000

Then call the product API and return matching products.

The user should never see API details.

Module 1: API-Based AI Assistant

Purpose

This module answers questions using live API data.

Examples:

Product Search

User: "Show red shirts under ₹1000"

AI extracts:

```
{ "intent": "product_search", "category": "shirt", "color": "red", "price_max": 1000 }
```

API Call:

```
GET /products?category=shirt&color=red&price_max=1000
```

Response:

```
[ { "name": "Red Cotton Shirt", "price": 799 } ]
```

AI Response:

"I found 5 red shirts under ₹1000. The lowest priced option is ₹799."

Order Tracking

User: "Track my order 1234"

AI extracts:

```
{ "intent": "track_order", "order_id": "1234" }
```

API Call:

GET /orders/1234

AI Response:

"Your order is currently out for delivery."

Appointment Booking

User: "Book an appointment tomorrow at 3 PM"

AI extracts:

```
{ "intent": "book_appointment", "date": "tomorrow", "time": "3 PM" }
```

API Call:

POST /appointments

AI Response:

"Your appointment has been booked successfully."

Important Requirement

The AI should never directly search the database.

Everything should happen through configured APIs.

Benefits:

- Secure
 - Flexible
 - Reusable
 - Works with any system
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Module 2: Website Knowledge Assistant

Purpose

This module answers questions using website content.

Sources:

- Home page
 - About page
 - FAQ page
 - Services page
 - Blogs
 - Policies
 - Uploaded documents
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Example

Website Content:

"We provide web development, mobile app development and cloud consulting services."

User:

"What services do you offer?"

Process:

1. Search relevant content.
2. Send content to AI.
3. Generate answer.

Response:

"We provide web development, mobile application development and cloud consulting services."

Knowledge Base Architecture

Step 1: Crawl Website

Extract:

- URL
- Page Title
- Content

Store in database.

Step 2: Chunk Content

Large pages should be divided into smaller pieces.

Example:

Page contains 5000 words.

Split into:

Chunk 1 Chunk 2 Chunk 3 ...

This improves AI search quality.

Step 3: Generate Embeddings

Convert text into vectors.

Example:

"Refund policy"

becomes

[0.23, 0.67, 0.12 ...]

Store vectors in pgvector.

Step 4: Semantic Search

User asks:

"Can I return a purchased item?"

Even if website contains:

"Refund Policy"

AI should still find the correct information.

This is the power of embeddings.

Hybrid Decision System

This is the most important part of the project.

The chatbot should decide:

Question: "Show red shirts"

Use: API

Question: "What is your refund policy?"

Use: Knowledge Base

Question: "Track my order"

Use: API

Question: "What services do you provide?"

Use: Knowledge Base

The AI should automatically choose the correct source.

Recommended Flow

User Message



Intent Detection



Decision Engine



API OR Knowledge Base



AI Response

↓

User

Admin Dashboard Features

The website owner should be able to:

- Create chatbot
 - Add website URL
 - Add API endpoints
 - Upload PDF files
 - Configure chatbot appearance
 - View conversations
 - View analytics
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Chat Widget Features

Required:

- Floating chat icon
- Mobile responsive
- Conversation history
- Typing indicator
- Dark mode
- Multi-language support

Optional:

- Voice input
 - Voice output
 - WhatsApp integration
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Future AI Features

AI Product Recommendations

User:

"I need a shirt for office use under ₹1500"

AI should recommend products even if the user doesn't specify exact filters.

Lead Generation

User:

"I am interested in your services"

Bot should ask:

"Can I have your name and email?"

Store lead in CRM.

Auto API Discovery

If website provides:

/swagger /openapi.json /redoc

The system should automatically detect available APIs and generate tools for the chatbot.

This can become a premium feature.

Development Priority

Phase 1:

- Chat widget
- Website crawler
- PDF ingestion
- Embeddings
- Knowledge search
- Product API integration

Phase 2:

- Hybrid AI routing
- Multiple APIs

- Authentication support
- Analytics

Phase 3:

- Action APIs
- Order tracking
- Booking systems
- WhatsApp
- Voice assistant

Success Criteria

The chatbot should be capable of answering:

1. Questions from website content.
2. Questions from uploaded documents.
3. Product search queries.
4. Live API-based queries.
5. Action requests such as booking and tracking.

The user should feel like they are interacting with a knowledgeable employee rather than a search box.